SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA04 **COURSE CODE:** Operating Systems

COURSE OUTCOME COVERED

CO1: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments.*

QUESTIONS: 1

Total Marks:100

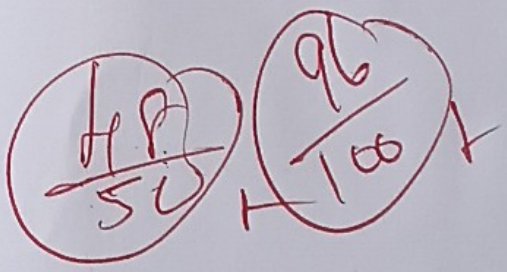
Annexure A

Technical Presentation

Code of Conduct:

I Ramya·M (192126014) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Technical Presentation [Any one]

Question 1
Suppose an Operating System is used in a banking system handling thousands of transactions simultaneously.
Prepare a technical presentation explaining:
Types of OS suitable for this scenario
OS structure used
Role of system calls in transaction processing **(BL3)**

Question 2
Consider a multi-user Linux-based server environment supporting file sharing and device access.
Prepare a technical presentation covering:
OS functions involved in resource management
File and device system calls
Interaction between user processes and kernel **(BL3)**

Question 3
Assume a real-time embedded system used in traffic signal control.
Prepare a technical presentation explaining:
Type of OS used and justification
OS structure and components
System calls used for process control and timing **(BL3)**

Question 4
A company uses a distributed computing system across multiple locations.
Prepare a technical presentation discussing:
Type of OS and its characteristics
Architecture of distributed OS
System calls used for communication and coordination **(BL3)**

Question 5
Consider an OS managing process scheduling, memory allocation, and I/O operations in a modern computer.
Prepare a technical presentation including:
Key OS functions and their importance
Interaction between OS components
Examples of system calls for each function **(BL3)**

RUBRICS FOR CALCULATION

Technical Presentation					
Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
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CSA0411-Operating System

Key OS Functions in a Modern Computer

Process scheduling, memory allocation and I/O operations

Submitted By:
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Submitted To:
Dr. Ria sudheen H
(Associate Prof.)

Key OS Functions and Why They Matter

1

Process Management

Creates, schedules and terminates processes; decides who gets the CPU.

Importance: Keeps the CPU busy and every program responsive.

2

Memory Management

Allocates and frees memory; maps virtual to physical addresses.

Importance: Isolates programs and lets more run than RAM alone allows.

3

I/O Management

Controls devices through drivers, buffers and interrupts.

Importance: Gives programs one simple interface to many different devices.

Process Management and Scheduling

The scheduler picks the next ready process to run on the CPU.

- FCFS: simple, runs in arrival order.
- Round Robin: fixed time slice, fair for interactive use.
- Priority: urgent tasks run first.

A context switch saves one process state and loads another.

Create a process	`fork()`
Load a new program	`exec()`
Wait for a child	`wait()`
Terminate	`exit()`

Memory Allocation and I/O Operations

Memory management

- Paging splits memory into fixed-size frames
- Virtual memory gives each process its own address space
- Page faults load data from disk on demand

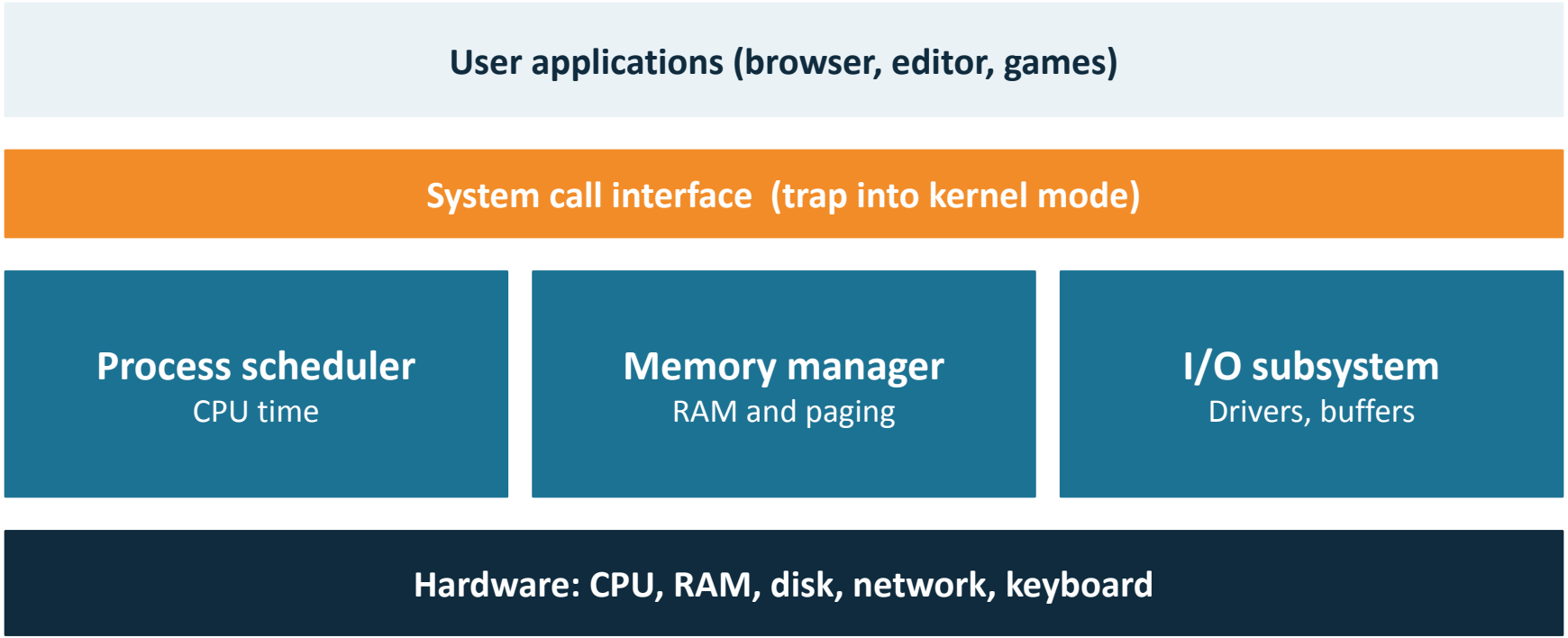
`malloc()` / `brk()` `mmap()`

I/O management

- Device drivers hide hardware differences
- Buffering and caching smooth speed gaps
- Interrupts signal that a device is ready

`open()` `read()` `write()` `close()`

How the OS Components Interact



Summary: Functions and System Calls

Function	Importance	Example system calls
Process scheduling	Fair, efficient CPU use	`fork(), exec(), wait(), exit()`
Memory allocation	Protection and efficient RAM use	`brk(), mmap(), munmap()`
I/O operations	Uniform device access	`open(), read(), write(), close()`

Conclusion: the OS coordinates the CPU, memory and devices through the kernel, and applications reach these services only via system calls.